

Example exercise 3

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15.55

Determine whether the following two CCS expressions

$$a.(b.Nil + c.Nil) \quad \text{and} \quad a.(b.Nil + \tau.c.Nil)$$

are:

- strongly bisimilar?
- weakly bisimilar?

Solution: The two expressions are neither strong nor weakly bisimilar.

Let

$$A = a.(b.Nil + c.Nil) \quad \text{and}$$

$$B = a.(b.Nil + \tau.c.Nil)$$

The LTS generated by A and B are as follows:

$$A \xrightarrow{a} b.Nil + c.Nil$$

$\begin{array}{c} b \searrow \\ \quad \quad \quad \swarrow c \\ \quad \quad \quad Nil \end{array}$

$$B \xrightarrow{a} b.Nil + \tau.c.Nil$$

$\begin{array}{ccc} & b \searrow & \swarrow \tau \\ Nil & & c.Nil \\ & \xleftarrow{c} & \end{array}$

There are different (but equivalent) ways to show that $A \not\sim B$ and $A \not\approx B$.

- Providing a distinguishing formula:

$$A \not\models \langle a \rangle \langle \tau \rangle tt$$

$$B \models \langle a \rangle \langle \tau \rangle tt$$

Then, by Hennessy-Milner theorem, we know that

$$A \not\sim B.$$

- By playing a (weak) bisimilarity game.

To show that $A \not\sim B$ we provide a universal winning strategy for **attacker**, starting from the configuration (A, B) .

→ From (A, B)

Attacker picks the transition $B \xrightarrow{a} b.\text{Nil} + \tau.c.\text{Nil}$.

Then **defender** has to reply with $A \xrightarrow{a} b.\text{Nil} + c.\text{Nil}$.

Hence, the next configuration is $(b.\text{Nil} + c.\text{Nil}, b.\text{Nil} + \tau.c.\text{Nil})$.

→ From $(b.\text{Nil} + c.\text{Nil}, b.\text{Nil} + \tau.c.\text{Nil})$

Attacker picks $b.\text{Nil} + \tau.c.\text{Nil} \xrightarrow{\tau} c.\text{Nil}$

Then **defender** is forced to reply with $b.\text{Nil} + c.\text{Nil} \xRightarrow{\tau} b.\text{Nil} + c.\text{Nil}$ (i.e., does not make any transition).

The next configuration is $(b.\text{Nil} + c.\text{Nil}, c.\text{Nil})$.

→ From $(b.\text{Nil} + c.\text{Nil}, c.\text{Nil})$

Attacker wins by picking $b.\text{Nil} + c.\text{Nil} \xrightarrow{b} \text{Nil}$.

Indeed **defender** has no available b -transitions.